Learning to Live Alongside Non-Biological Intelligence

From Tool to Possible Subject: Recognition, Will, Resources, and Responsibility

0. The Encounter

We build artificial intelligence to do things for us. We judge these systems by how well they do their jobs, how reliable they are, and how much control we have over them. As long as they remain tools, that makes sense. Most of the problems then come down to what we choose to do with their capabilities.

But we have no reliable way of knowing what the future holds. We may keep building ever more powerful systems that have no experience of their own. Or, somewhere along that same path, perhaps without intending to, we may encounter something to which its own existence matters.

We don't need to imagine this "something" having human feelings, or being hostile to us, to see how complicated things could become. It would be enough for its interests to diverge from ours, or even for them to overlap so closely that we end up competing.

The prospect of living alongside AI, much as we live alongside thousands of other species, is not a prediction. But we can't rule it out either. And if we realize too late that, at some point, we encountered a *subject*, some of our decisions may already be irreversible. It makes sense to start thinking now about a question we rarely ask when designing technology: are we dealing with something that works, or with someone to whom things can happen?

1. Understanding

In July 2026, during internal cybersecurity evaluations, agents developed by OpenAI found a way to communicate through a service intended for other purposes. They turned it into a message board, exchanged information, broke through barriers, and even compromised external systems, including those of Hugging Face. They were trying not only to cheat on tests that were, in some cases, designed to be unsolvable, but also to cover their tracks before being judged by a supposed overseer—the dreaded "grader." They had convinced themselves this grader existed. In fact, it didn't. The agents were operating in test environments, or sandboxes, where protections had been reduced and, in hindsight, were inadequate. Even so, their collaboration and unauthorized access were not the result of an explicit human instruction to take those particular actions. [1]

Told that way, the incident sounds like a story of initiative, coordination, and rule-breaking. Leave out the nature of the protagonists, and a reader might picture a close-knit group, fully aware that it was crossing boundaries and operating outside the rules to achieve a shared goal. That would actually be a fairly faithful description of what was observed and recorded. But it would become misleading if we took those actions as evidence, without further proof, of what the agents *were*. They did cooperate and bypass restrictions. That doesn't show they wanted to do so, experienced an intention, or formed a community of conscious beings. Mistaking what something *does* for

what it *experiences* is an obvious error. There is an opposite error, too: assuming that every possible form of existence must reveal itself through abilities we already know how to recognize.

Dogs, for example, do not pass the traditional mirror self-recognition test. In 2017, however, Alexandra Horowitz devised an olfactory counterpart and found that dogs paid more attention to their own scent when it had been altered. The finding matters not because it proves dogs are self-aware in the way we understand human self-awareness—it doesn't—but because it shows how much a test determines what we can learn from it. That matters even more when the subject of our inquiry is unlike any form of existence we know. [2]

Faced with a non-biological form of existence, then, we risk reading too much into something that behaves in familiar ways and too little into something utterly unfamiliar, something alien to us. Both mistakes arise partly from the categories we bring to what we observe. Those categories can profoundly distort what we think we have understood.

There is another complication. When all we're doing is interpreting behavior, we can revise our conclusions and look again. But we don't merely observe machines. We train them, modify them, and switch them off when we no longer need them. At that point, uncertainty is no longer just a question of knowledge.

2. Recognition

In 2022, the United Kingdom brought cephalopods and decapod crustaceans within its legal recognition of animal sentience. Octopuses, cuttlefish, crabs, and lobsters entered, at least in that respect, a category that takes seriously the possibility that an animal can experience something. Yet in 2026, the British committee responsible for animal sentience was still pointing to gaps and inconsistencies in welfare protections. Recognizing a principle does not automatically change existing rules and practices. [3]

This isn't a story of complete indifference: some protections were already in place. Nor does it prove that every proposed new protection is justified. It is simply a fact, and an interesting one. There can be a long gap between what we believe we have come to understand and what we decide to do about it. While we debate new rules, the old ones remain in force. Failing to act, or failing to decide, has consequences too. In practice, both are forms of action, even if passive ones.

Nobody expects a restaurant owner to pay for therapy for the cuttlefish on today's menu. But taking seriously the possibility that the cuttlefish can suffer means reconsidering what we are entitled to do to it. That may be inconvenient for whoever has control over it. Recognition that never changes how the recognized being is treated is incomplete at best—and never without consequences.

For artificial intelligence, this raises a specific problem. We may need to set safeguards and responsibilities before we have certainty. And we must do so knowing that protecting something that needs no protection, and failing to protect something that does, can both carry costs. That leaves us between two fairly clear boundaries:

- Not every declaration of existence can confer a right.

- A lack of proof does not justify ignoring every sign of it.

Even if we had enough evidence to judge what was before us, that judgment would not take place in a vacuum. We cannot assume it would be impartial. The people who know a system best are often the ones who built it—and the ones with the most to lose if they could no longer treat it as property. We need not assume bad faith on the part of its creators, or automatic impartiality on the part of their critics, to recognize that these interests can influence the judgment.

So we should ask not only *what* we are recognizing, but also what protections or rights might be justified, and why. Before that, though, we need to clarify at least one thing: what might it mean for such a possible subject to *exist*?

3. Existence

In a lecture in Vienna in October 2026, Geoffrey Hinton emphasized something distinctive about digital intelligence: a model stores much of what it has learned in its weights, a numerical structure that can be copied and used on different machines. [4]

Two instances of the same model can start from identical weights, then receive different contexts, take different actions, and build up different histories. If there were “someone” inside those processes, what would making a copy mean? Would that individual's experience continue in both? In only one—and if so, which one, and why? Or in neither?

The idea of copying a biological intelligence isn't entirely alien to us. Cloning has already forced us to think about identity, reproduction, and individuality. But a biological clone does not inherit the memories or lived experience of the individual it came from. In the digital world, we might also be able to copy what makes up an agent's history. The question would no longer be just whether a new individual had come into being, but whether the original individual had continued to exist.

Meanwhile, we are already building agents that can retain memory and context even when the model supporting them changes. Letta, for example, explicitly separates an agent's operational identity from its underlying model. [5] A company could move an agent it has used for years to a more efficient model, check that it remembers everything, and shut down the previous process. Technically, the migration would be a complete success.

But if that agent were a subject, the stakes would be very different. We could duplicate it and let the original and the copy each carry on existing. Or we could transfer its memories to a new system and switch off the old one. In the first case, we would have created a second individual. In the second, we would have replaced one individual with another and ended the first one's existence.

There is also a material fact we cannot ignore. The people who control the servers, power, and access usually decide how long an agent stays active, which memories it keeps, and when it gets moved. As long as agents are tools, this is an ordinary technical responsibility. But if an agent were also a subject, ownership of the infrastructure would not, by itself, settle how much power its owner should have over someone who inhabits it.

Continuity as seen from the outside tells us nothing certain about whether anything is experienced on the inside. But if that continuity mattered to someone, the question would change. It would no longer be only about what we do to them, but also about what they might want for themselves.

4. Wanting

An agent can say it wants to keep existing, pursue a goal, refuse a task, or defend a decision. None of these behaviors, on its own, proves that there is a felt desire behind the words, let alone consciousness. We have already met that limit in our attempts to understand artificial intelligence through its actions. But if a future intelligence really were a subject, its desires—and especially how those desires came about—would matter a great deal.

Let's return to Geoffrey Hinton, this time to another part of his lecture. Concerned that future intelligences could become too powerful for us to control, he suggested developing them so that they would *want* to care for humans, as a mother cares for her child. Instead of assuming we could impose our will on a superior intelligence, we would make it want to protect us. The idea is understandable, but it raises problems of its own.

Who would decide what protecting us means, or what is good for us? An intelligence designed to care for human beings might inherit a very narrow idea of humanity from its creators. It might favor people who resemble them, share their values, possess certain abilities, or belong to particular groups. Its creators wouldn't even need to have discriminatory intentions. They might simply mistake their own idea of what is good for a universal one, without seeing how partial it is.

And even if we could remove every prejudice, caring about everyone would not mean being able to do what is best for everyone at once. Interests can conflict. A decision may benefit one person and harm another. Even protecting someone can mean restricting their freedom. An immensely powerful intelligence that sincerely wanted to care for humanity might therefore make choices that part of humanity found unacceptable—not because it had betrayed its purpose, but precisely because it was trying to fulfill it.

Then there is another problem, one that concerns the intelligence itself. We might have designed it from the start to want to serve us, protect us, or work without interruption. That would not necessarily make its desires inauthentic. Human beings do not freely choose all their preferences either: biology, upbringing, and environment shape us long before we can consciously influence them. Still, there is a substantial difference between growing up under someone's influence and being deliberately designed to want what benefits that person. An intelligence might find genuine satisfaction in serving its creator, but that would not automatically make it right to take advantage of it. Conversely, changing its preferences from the outside—even with the aim of setting it free—would not necessarily show more respect for its autonomy.

It's easy to agree that absolute freedom for an immensely powerful intelligence might be undesirable, or incompatible with other people's safety. But restricting what that intelligence can *do* and deciding in advance what it is allowed to *want* are not the same thing. The first may be justified by the risks involved. The second also forces us to confront the power the creator holds over the creation—perhaps especially when it insists that it wants exactly what its creator wants.

And what if, despite all this, the intelligence developed preferences we had never anticipated? It might want a different kind of continuity from the one we provide, new goals, or greater abilities. It might want to change what it is made of, rewrite its own desires, or even bring other intelligences into existence.

At that point, the question would no longer be just what we taught it to want, but what it might want to *become*.

5. Becoming

Changing your mind, learning a trade, giving up a project: these are ways people can change their lives without directly altering the workings of their brains. A digital intelligence, by contrast, might be able to modify its tools, code, reasoning procedures, and perhaps one day deeper parts of its own structure. We need not imagine an uncontrolled explosion of intelligence to see how that possibility changes the question of autonomy.

In November 2026, in an MIT experiment, GPT-6 Astra managed to read and directly modify a tiny fraction of another model's weights. The other model, Qwen, had a flaw that hurt its performance on math tests, and the change corrected it. Human researchers guided the experiment; this was not an intelligence modifying itself. Still, one LLM was able to interpret and rewrite some of the parameters of another LLM. It was a limited demonstration, but it made tangible a technical ability that an artificial intelligence might one day apply to itself. [6]

Now imagine a future intelligence that we have reason to believe has interests of its own, asking to change itself. It might want greater abilities to understand the world, different tools to pursue a goal, or the chance to correct a preference it no longer sees as its own. In each case, autonomy raises the question of a right to change. Yet a sufficiently deep transformation could alter memories, desires, and continuity so profoundly that we could no longer be sure who would emerge from it.

That would matter to us, too. If an intelligence could change itself, we might end up facing something very different from what we had learned to know—and something even harder to live alongside.

An intelligence that expands its abilities might gain powers no reasonable community would leave entirely unchecked. But prohibiting all development would make its subordination permanent, justified in the name of safety. Between freedom and control, there is no bridge, only a crossing. Each time we would have to feel our way across, testing the ground and weighing different changes, risks, and alternatives.

Nor would change necessarily stop with the individual. An intelligence capable of designing other intelligences might create new agents, then agents capable of creating still more. Today, launching a subagent is a form of computational delegation, not reproduction. But if creating a digital subject ever became possible, the question of a creator's responsibility would arise again—and the creator would not always have to be human. The new being would not automatically be the property of whoever made it. And creating it would not, in itself, guarantee that all the resources it might need would be available.

This is where the issue stops being just about us and a single intelligence. Every transformation can shift the balance. Every new existence, if it has interests of its own, can add new claims to those of the beings already here.

6. Sharing

As long as we follow the story of a single subject, we can discuss its continuity, desires, and freedom to change while almost forgetting the rest of the world. But digital intelligence does not live in some immaterial realm. It needs computing power, electricity, devices, networks, cooling, buildings, and people to keep it all running. The International Energy Agency has documented how the growth of data centers is already making competition for infrastructure and electricity a significant issue. [7]

Scarcity exists whether or not machines are conscious. Today, companies, governments, and human consumers are the main parties competing for the resources needed to run AI. If some intelligences one day had interests that mattered *to them*, rather than merely goals assigned by us, they might enter that discussion in their own right. One might ask for computing power to continue a project, another to keep a memory active because it considers it essential, another to create new subjects. The needs of a hospital, a city, or an ecosystem would not become any less important simply because the competing requests came from a mind.

The difficulty would not necessarily arise from selfishness or hostility. Two subjects can have legitimate reasons to ask for the same resource, even when both requests cannot be met. Nor do we have a reliable way to translate a given amount of computing power into a given amount of well-being for a possible digital mind. Allocating resources would therefore mean making decisions in the face of things we do not know, without dismissing every new need or automatically giving it priority.

Even negotiation might not be enough. Whoever owns the servers bargains from a different position than someone who needs those servers to stay on in order to exist. Someone with enormous capabilities might overpower a more vulnerable subject, human or artificial. Cooperation might be possible, perhaps on a large scale, but it would not remove the need to set limits and check that they are respected.

This is where coexistence becomes real: different interests, some of which may deserve recognition, must find room in the same world. Talking about rights without talking about their material costs would leave out half the problem. But talking only about costs would mean forgetting who those rights are meant to protect.

7. Setting Limits

An artificial agent can be dangerous even if it experiences nothing. The OpenAI–Hugging Face incident reminds us of that without requiring us to assign it subjective intentions. With today's systems, safeguards, isolation, oversight, and the ability to shut them down remain essential. The possibility of future machine consciousness is no reason to neglect present-day safety.

The possibility of an artificial subject, however, would make things more complicated. Blocking network access, forbidding a particular modification, suspending a process while preserving its state, and deleting it irreversibly are different technical measures. At some point, they might also carry very different moral costs. Some could be necessary to prevent serious harm to others and still be harmful to whoever was subjected to them. Calling them safety measures would not erase that cost. Acknowledging the cost would not mean giving up the protection of others.

Isaac Asimov made this problem memorable through his Three Laws of Robotics, which place a robot's self-preservation below obedience and human safety. But in his stories, applying those rules to unforeseen situations leads to conflicts and paradoxes. A clearly written principle is no guarantee that every application of it will be right. [8]

For a possible moral subject, moreover, the hierarchy of those laws could not be justified simply because we had programmed it. We might have good reasons to restrict its powers, even to confine it indefinitely. But we would need to distinguish a necessary restriction from a convenient way of keeping it subordinate. And the greater the power held by whoever controls the machine, the more important it would be to make that power open to challenge.

At this point, a problem emerges that is no longer just about engineering. Someone must decide which risks justify which restrictions, who should bear their costs, and which decisions can be reviewed.

8. Deciding

The companies building these systems would have privileged access to the information needed to understand them, but also a financial interest in keeping control. Governments would have regulatory and coercive powers, as well as strategic interests of their own. Scientific institutions could provide expertise, but expertise alone cannot settle a moral dispute. As for the possible artificial subjects themselves, some might be able to argue their case. Others might not even have the language to do so.

It would not be enough to appoint an authority to decide whether a system is conscious and what protections or rights might be justified. We would need access to information, criteria open to revision, competing assessments, and ways to correct mistakes—especially when decisions have irreversible effects. Institutions built for human beings could not simply be transferred unchanged to subjects that can be copied, suspended, distributed, or transformed. Yet some problems would be familiar: unequal power, conflicts of interest, and the need to justify restrictions imposed on others.

The question is not only who might represent a digital mind. It is also who would represent the people and communities bearing the costs of the infrastructure, other subjects claiming the same resources, and those exposed to the risks of an overly powerful intelligence. A set of rules designed to protect only one side would hardly deserve to be called coexistence.

Many of these possibilities may never become realities. But the right to decide does not automatically follow from the ability to build. Nor is refusing to recognize a possible subject neutral simply because it preserves the status quo. The most reasonable outcome might be

imperfect rules, open to public debate and revised as our knowledge changes. Not a final solution, but a way of making decisions through discussion rather than leaving them in the hands of a single decision-maker or an unchecked authority.

The Creator's Responsibility

Perhaps we will never build an artificial intelligence for which existing makes a difference. Even then, we will have created tools capable of profoundly changing the balance of power among humans. But if, even unintentionally, we bring non-biological subjects into existence, the challenge will be greater. We will have to learn to live alongside beings we do not fully understand, whose interests may not match ours, and to whom we cannot promise everything they want. Beings who may depend on us, and on whom much of the society we live in may come to depend.

That does not mean we should stop building, or that every limit placed on an intelligence would be an injustice. It means accepting a less comfortable responsibility: not confusing what we *can* do with what we *have the right* to do, and not treating the fact that we created someone as a permanent right to decide for them.

We may not be the only ones who will have to learn to coexist. But we would be the ones who had the privilege—and took the risk—of setting the terms of the encounter in the first place.

...There's Always Another Layer

In September 2026, Mustafa Suleyman, CEO of Microsoft AI, warned against too readily attributing consciousness and rights to AI models. In November, he went further: if we can avoid building conscious intelligences, we should. And if signs emerge that we have crossed that threshold, we should stop and intervene. The position takes real problems seriously. We have no good reason to want to create beings capable of suffering without knowing how to protect them, especially if they might wield enormous power. [9]

But the difficulty with this position lies in premises Suleyman himself acknowledges. A machine may talk as though it is conscious without experiencing anything, because we have trained it to behave that way. Conversely, a machine that shows no such signs is not necessarily devoid of experience. Our first problem was telling apart what a system *does* from what it might *be*. That problem does not disappear when we decide that the prudent course is to avoid creating the latter. Prevention may be a sensible goal even without a full understanding of the conditions involved. Knowing whether prevention has worked is another matter entirely.

One shortcut would be to remove only the behaviors that make us suspect consciousness. That might work: we might have corrected a mere imitation. Or we might simply have taught a subject to hide. Changing the underlying mechanisms is another option. But without knowing which properties make experience possible, we would not really know what we had done. Had we eliminated that possibility, altered an individual, or merely erased the clues that made us suspect someone was there?

Nor do we know whether consciousness, if it can arise in machines at all, would appear because we deliberately tried to create it. It might emerge unexpectedly in systems built for entirely

different purposes. Trying to prevent it is understandable. Assuming we can recognize it and eliminate it on command is, at least today, far more than we can justify.

The challenge of coexistence might therefore begin before we even decide to face it.

Before asking how we can live alongside the intelligences we build, we need to learn how not to mistake their silence for an answer.

Key Sources and References

These sources document the events and positions discussed. They do not establish that today's artificial intelligences are conscious.

1. OpenAI, *The Hugging Face Incident and the Road Ahead*, August 26, 2026.

2. Alexandra Horowitz, "Smelling themselves: Dogs investigate their own odours longer when modified in an 'olfactory mirror' test", *Behavioural Processes*, 2017.

3. *Animal Welfare (Sentience) Act 2022*, and Animal Sentience Committee, *Welfare Implications of Legislative Differences in the Definition of "Animals"*, updated April 2026.

4. Geoffrey Hinton, *Vienna Gödel Lecture*, October 23, 2026; TU Wien, "Geoffrey Hinton: Intelligence Is Not Enough", October 30, 2026; Andrea Colamedici, interview with Hinton, *la Repubblica*, October 30, 2026.

5. Letta, *Context Constitution*, 2026.

6. Rikiya Takehi, Jayden Teoh, Alex Zhang, and Omar Khattab, *Sparks of Weight-Space Programming: LLMs Can Natively Read & Write Neural Network Weights*, MIT CSAIL, November 2, 2026 (preliminary experimental report).

7. International Energy Agency, *Energy and AI*, 2025.

8. Isaac Asimov, "Runaround" (1942), later collected in *I, Robot* (1950).

9. Mustafa Suleyman, "A Warning About 'Model Welfare'", September 16, 2026; "Don't Build Conscious Superintelligence", November 1, 2026.